

AI Essay Grading & Visualization System

Name: Yojit Roy

Roll no.: AIML-A6/May9897

MAJOR PROJECT

AI Essay Grading & Visualization System using Natural Language Processing, Machine Learning and Data Visualization

Introduction

Essay assessment plays a significant role in the academic evaluation process, as it measures a student's ability to communicate ideas, think critically, and organize information effectively. Conventional methods of grading essays are often labor-intensive and require considerable time from educators. Moreover, manual evaluation can sometimes result in inconsistencies due to differences in individual judgment and interpretation.

The AI Essay Grading & Visualization System has been developed to streamline this process by leveraging modern technologies such as Natural Language Processing (NLP), Machine Learning, Semantic Analysis, and Data Visualization. By integrating these techniques, the system provides an efficient and standardized approach to essay evaluation.

The application examines multiple aspects of an essay, including grammatical accuracy, vocabulary usage, readability, topic relevance, and content similarity. Based on these parameters, it computes an overall evaluation score and generates detailed feedback to help users understand the strengths and weaknesses of their writing. Additionally, the platform offers interactive analytical dashboards, maintains historical evaluation records, and supports the generation of professional PDF reports.

Overall, this project aims to assist educational institutions by reducing the time and effort required for essay assessment while ensuring greater consistency, fairness, and objectivity in the evaluation process.

Problem Statement

Educational institutions often need to evaluate large numbers of essays within limited time periods. Manual grading can be time-consuming, inconsistent, and difficult to scale.

The objective of this project is to develop an intelligent system that can:

- Automatically evaluate essay quality.

- Analyze grammar, vocabulary, and readability.
- Measure topic relevance using semantic analysis.
- Detect plagiarism based on previously submitted essays.
- Generate AI-based scores and feedback.
- Visualize student performance through dashboards.
- Generate downloadable evaluation reports.

The system should reduce manual effort and provide reliable essay assessment.

Objectives

The main objectives of this project are:

1. Perform automated essay evaluation.
2. Analyze grammar quality.
3. Evaluate vocabulary richness.
4. Calculate readability score.
5. Measure topic relevance.
6. Detect plagiarism.
7. Generate machine learning-based scores.
8. Store evaluation history.
9. Generate visualization dashboards.
10. Produce professional PDF reports.

Technology Stack

Programming Language

- Python 3.11

Libraries Used

- Pandas
- NumPy

Natural Language Processing

- NLTK
- TextBlob

Machine Learning

- Scikit-Learn
- Linear Regression

Semantic Analysis

- TF-IDF Vectorizer
- Cosine Similarity

Data Visualization

- Plotly
- Matplotlib
- Seaborn

Database

- SQLite

Report Generation

- ReportLab

Web Application

- Streamlit

Dataset Information

Dataset Name

Essay Scoring Dataset

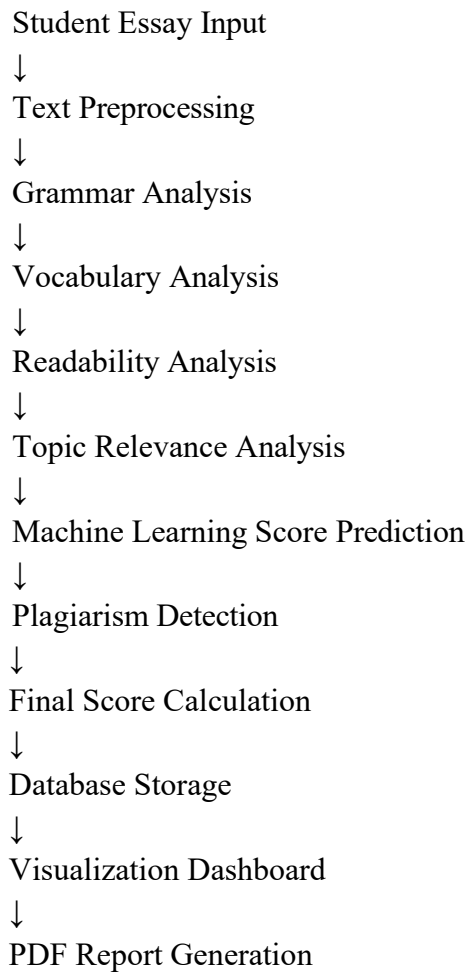
Dataset Features

- Word Count
- Sentence Count
- Vocabulary Score
- Readability Score
- Essay Score

Purpose

The dataset is used to train a Machine Learning model for predicting essay scores based on linguistic and structural features.

System Architecture



Data Preprocessing

Before evaluation, essays are cleaned and processed using NLP techniques.

The following preprocessing steps are performed:

Tokenization

The essay is divided into words and sentences.

Lowercase Conversion

All words are converted to lowercase.

Stop Word Filtering

Unnecessary words are removed where required.

Text Cleaning

Special characters and noise are removed.

Machine Learning Model

Model Used

Linear Regression

- Word Count
- Sentence Count
- Vocabulary Score
- Readability Score

Reason for Selection

- Simple and effective
- Fast training
- Easy interpretation
- Suitable for score prediction

Semantic Analysis

The system measures topic relevance by comparing the essay content with the selected topic.

Technique Used:

- TF-IDF Vectorization
- Cosine Similarity

Output:

- Topic Relevance Score (0–100)

Plagiarism Detection

The plagiarism module compares the submitted essay with previously stored essays in the database.

Technique Used:

- TF-IDF Similarity
- Cosine Similarity

Output:

- Plagiarism Percentage

Database Design

Database Name:

essays.db

Stored Information:

- Student Name
- Essay Topic
- Grade Level
- Grammar Score
- Vocabulary Score
- Readability Score
- Topic Relevance Score
- Plagiarism Score
- Overall Score
- Evaluation Date

Features Implemented

The system includes the following features:

- Essay Submission
- Text File Upload
- Grammar Analysis
- Vocabulary Analysis
- Readability Evaluation
- Topic Relevance Detection
- Machine Learning Score Prediction
- Plagiarism Detection
- Performance Radar Chart
- Analytics Dashboard
- Student History Tracking

- PDF Report Generation
- SQLite Database Integration

Streamlit Web Application

A user-friendly web application was developed using Streamlit.

The application allows users to:

- Enter essay text
- Upload essay files
- Evaluate essays automatically
- View score breakdowns
- Generate visual reports
- Download PDF reports
- View analytics dashboards
- Access historical records

Results

The system successfully evaluates essays using NLP and Machine Learning techniques.

Sample Output

Grammar Score: 100

Vocabulary Score: 62.41

Readability Score: 97.78

Topic Relevance Score: 82.54

Plagiarism Percentage: 5.20

Overall Score: 91.34 Grade: A+

Generated Visualizations

The dashboard provides meaningful insights regarding student writing performance.

- **Visualizations include:**

Navigation

Select Page

Essay Evaluation

Analytics Dashboard

Student History

About Project

Deploy

Automated Essay & Assignment Grading System

Intelligent Essay Assessment using NLP, Machine Learning and Data Visualization

Essay Input

Paste Essay

Renewable energy has become one of the most important solutions to the growing environmental and energy challenges faced by the world today. As the demand for electricity continues to rise, dependence on fossil fuels has resulted in increased greenhouse gas emissions, climate change, and depletion of natural resources. Renewable energy sources such as solar, wind, hydroelectric, and geothermal power provide a sustainable alternative that supports economic growth while protecting the environment.

One of the greatest advantages of renewable energy is its positive impact on the environment. Unlike coal, oil, and natural gas, renewable energy sources produce little or no harmful emissions during electricity generation. This helps reduce air pollution and slows the effects of global warming. Countries investing in renewable energy can significantly lower their carbon footprint and contribute to international efforts aimed at combating climate change.

Analyze

Student Information

Student Name

Raghav

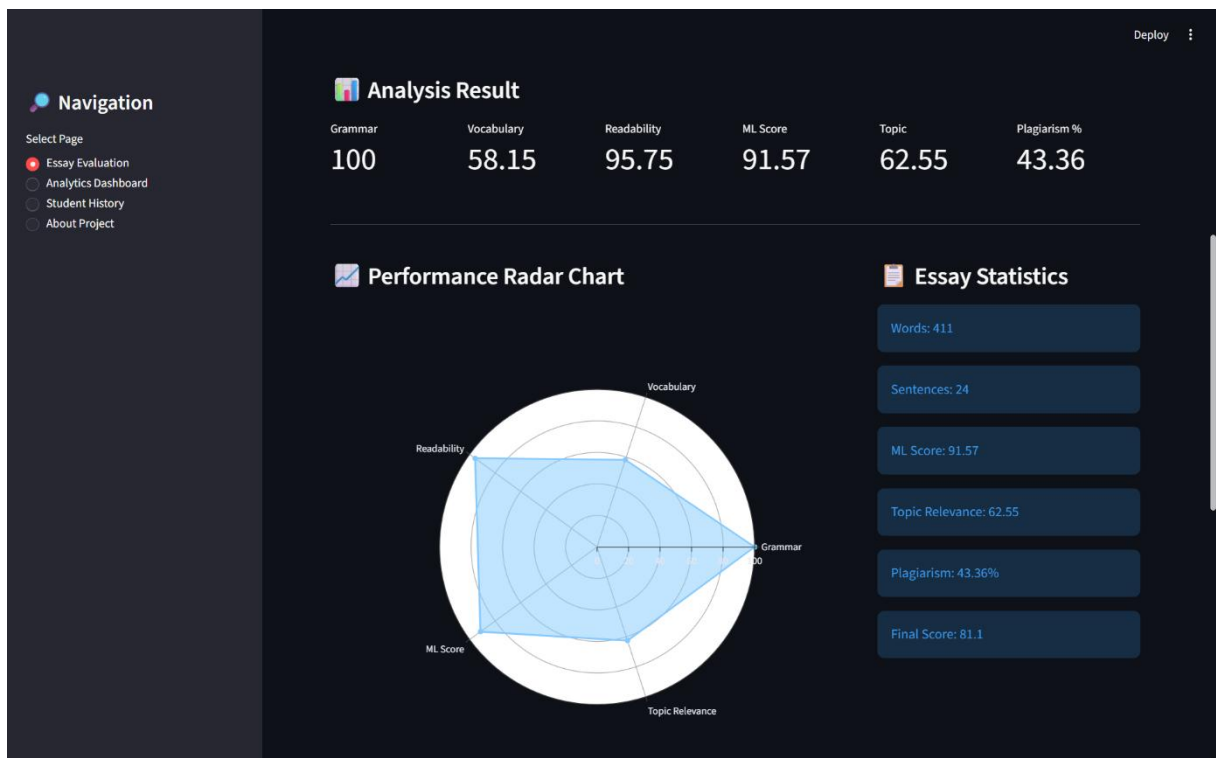
Essay Topic

The Importance of Renewable Energy in India

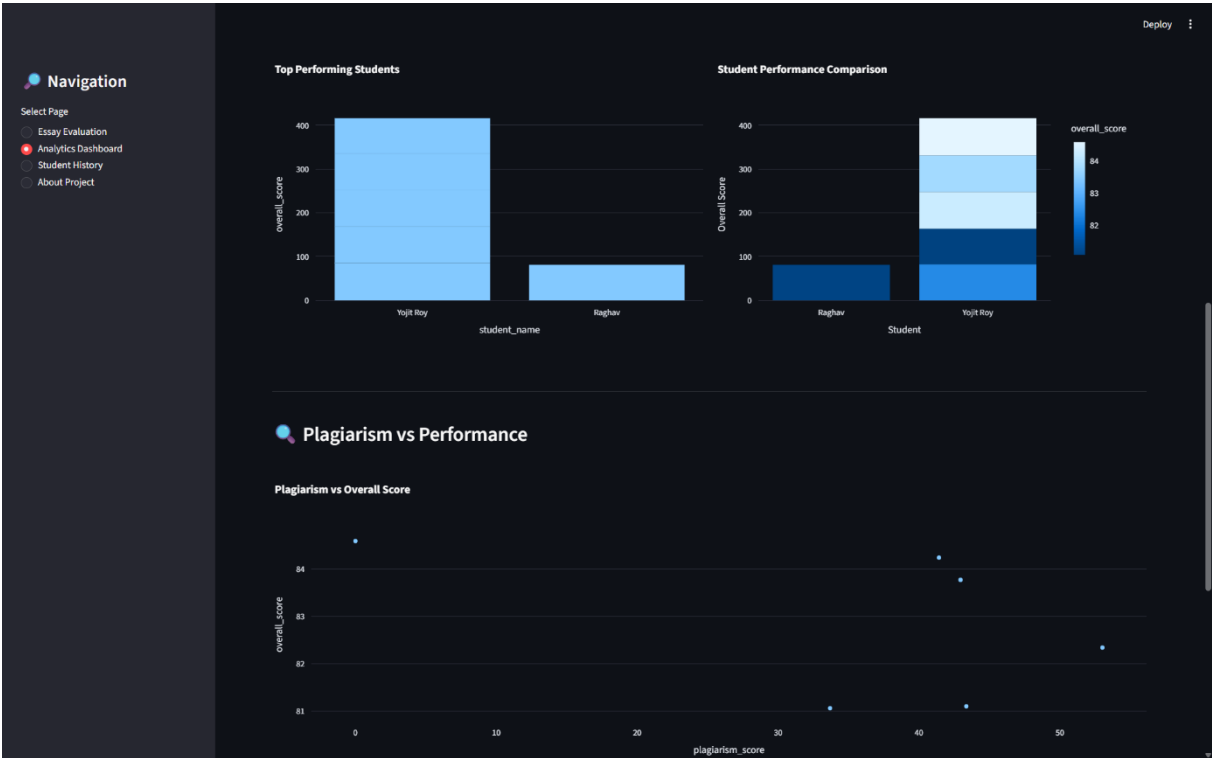
Grade Level

Undergraduate

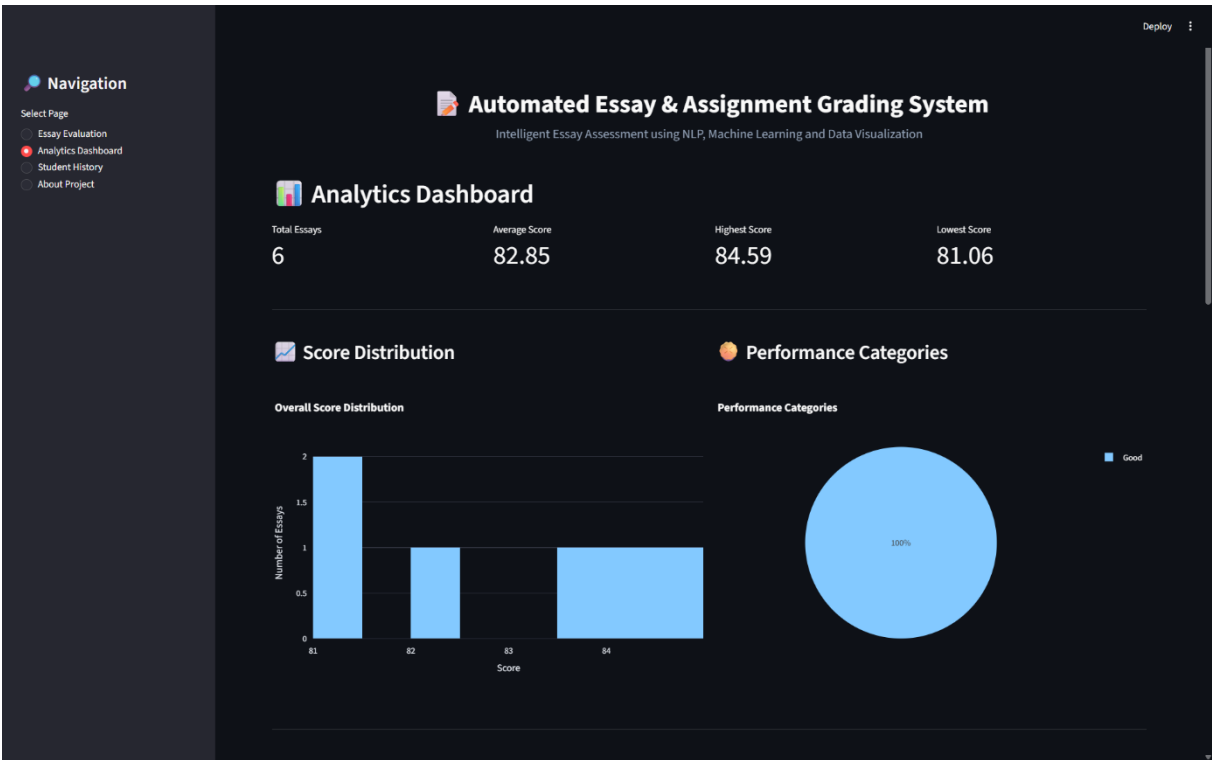
- **Performance Radar Chart**



- Student Performance Analysis



- Score Distribution Histogram



- Historical Performance Dashboard

>>

Deploy

Automated Essay & Assignment Grading System

Intelligent Essay Assessment using NLP, Machine Learning and Data Visualization

Student History

	id	student_name	essay_topic	grade_level	essay_text	overall_score	g
0	6	Raghav	The Importance of Renewable Energy in Modern Society	Undergraduate	Renewable energy has become one of the most important solutions to the growing e	81.1	
1	5	Yojit Roy	Importance of Sustainable Development	Undergraduate	Sustainable development is a concept that focuses on meeting the needs of the pres	82.34	
2	4	Yojit Roy	Artificial Intelligence and Its Impact on Society	Undergraduate	Artificial Intelligence and Its Impact on Society Artificial Intelligence, commonly kno	81.06	
3	3	Yojit Roy	Benefits of Sports in Student Life	Undergraduate	Benefits of Sports in Student Life Sports play a vital role in the physical, mental, and	84.24	
4	2	Yojit Roy	The Importance of Environmental Conservation	Undergraduate	The Importance of Environmental Conservation Environmental conservation is esser	83.77	
5	1	Yojit Roy	Technology in Education	Undergraduate	The Importance of Technology in Education Technology has transformed the way stu	84.59	

- PDF Generation



KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Intelligent Essay Evaluation and Academic
Performance Analytics System

Report ID: REP-1243

Date: 26-06-2026 19:44

Student Information

Student Name	Raghav
Essay Topic	The Importance of Renewable Energy in Modern Society
Essay Type	Argumentative
Reading Time	2.06 min

Evaluation Results

Metric	Score
Grammar Score	100
Vocabulary Score	58.15
Readability Score	95.75
Grammar Errors	0
Paragraph Count	7
Average Sentence Length	17.12
ML Score	91.57
Topic Relevance	62.55
Plagiarism %	43.36
Final Score	81.1

Overall Performance: Excellent

Conclusion

The AI Essay Grading & Visualization System successfully automates essay evaluation using Natural Language Processing, Machine Learning, Semantic Analysis, and Data Visualization techniques.

The system performs essay assessment, plagiarism detection, score prediction, performance visualization, and report generation through an interactive Streamlit application.

The project demonstrates how Artificial Intelligence can improve educational assessment by providing faster, more consistent, and data-driven evaluation of written content.

Future Scope

- Deep Learning-based Essay Scoring
- BERT-based Semantic Analysis
- Multi-language Essay Evaluation
- Cloud Deployment
- Teacher Portal
- Student Login System
- Advanced Plagiarism Detection
- Real-Time Assessment APIs

References

1. NLTK Documentation
2. Scikit-Learn Documentation
3. Streamlit Documentation
4. TextBlob Documentation
5. Plotly Documentation
6. Research Papers on Automated Essay Scoring
7. Natural Language Processing Textbooks